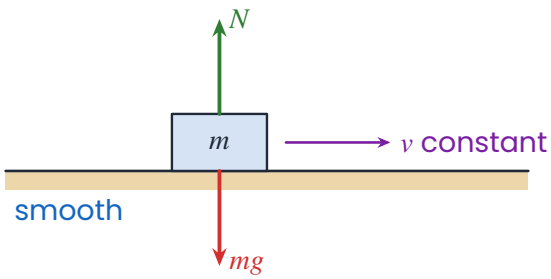
LAWS OF MOTION

Newton's Laws

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1 First Law • Inertia

$$\sum \vec{F} = 0 \iff \vec{v} = \text{constant}$$

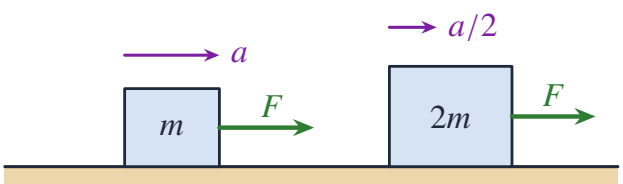


Holds in an inertial frame. Zero net force means no change in velocity, not no motion.

2 Second Law

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

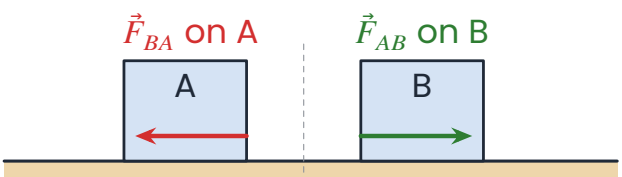
$$\vec{F}_{\text{net}} = m\vec{a} \quad (m \text{ constant})$$



Inertial frame; $\vec{a}$ is along the NET external force. Same F, double m: $a/2$.

3 Third Law

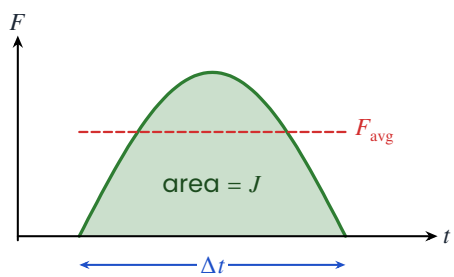
$$\vec{F}_{AB} = -\vec{F}_{BA}$$



Equal and opposite but on DIFFERENT bodies, so they never cancel each other.

4 Impulse

$$\vec{J} = \int \vec{F}\, dt = \Delta\vec{p} = \vec{F}_{\text{avg}}\, \Delta t$$



$\vec{F}$ = net force; J = area under the F–t graph. Longer contact time, smaller force.

LAWS OF MOTION

Momentum • Free Body Diagram

JEE & NEET PHYSICS

5 Conservation of Momentum

$$\sum \vec{F}_{\text{ext}} = 0 \Rightarrow \vec{p}_{\text{total}} = \text{const}$$

Recoil: $MV = mv$

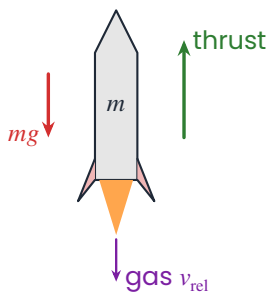


$p_{\text{total}} = 0$ before and after

Only when the net external force is zero. Gun and bullet start at rest, so they move opposite.

6 Rocket • Variable Mass

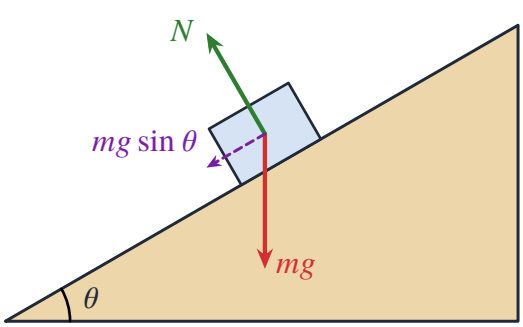
$$F_{\text{thrust}} = v_{\text{rel}} \frac{dm}{dt}, \qquad v = v_{\text{rel}} \ln \frac{m_0}{m}$$



v_{rel}: gas speed relative to the rocket. v formula: from rest, no gravity.

7 FBD: Smooth Incline

$$N = mg\cos\theta, \qquad a = g\sin\theta$$



Smooth fixed incline. Draw only the forces ON the body; $N \perp$ surface.

8 Quick Numerical: Impulse

A 0.2 kg ball hits a wall at 10 m/s and rebounds at 8 m/s. Contact time 0.02 s. Impulse and average force?

$$J = m(v + u) = 3.6 \text{ N s}$$

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{3.6}{0.02} = 180 \text{ N}$$

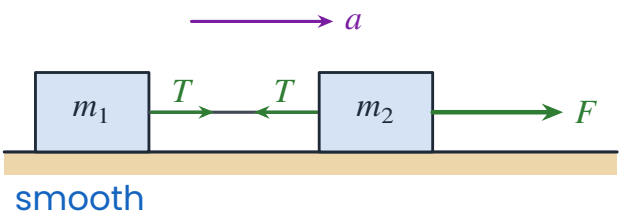
LAWS OF MOTION

Strings • Pulleys

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9 Connected Blocks

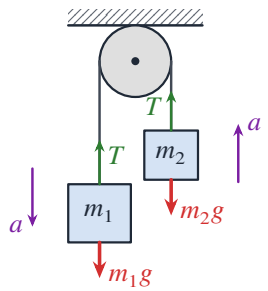
$$a = \frac{F}{m_1 + m_2}, \qquad T = \frac{m_1 F}{m_1 + m_2}$$



Smooth floor, light inextensible string; T = force on the block it drags.

10 Atwood Machine

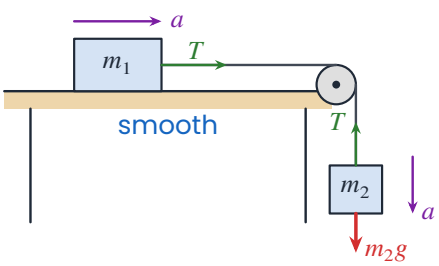
$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$



Light string, light frictionless pulley, $m_1 > m_2$. Same T on both sides.

11 Block on Table + Hanging Block

$$a = \frac{m_2 g}{m_1 + m_2}, \qquad T = \frac{m_1 m_2 g}{m_1 + m_2}$$



Smooth table, light string and frictionless pulley. $T < m_2 g$ while m_2 accelerates down.

12 Quick Numerical: Atwood

Atwood machine: $m_1 = 3$ kg, $m_2 = 2$ kg, light string and pulley, $g = 10$ m s^{-2}. Find a, T and the force on the support.

$$a = \frac{(3-2)10}{5} = 2 \text{ m s}^{-2}$$

$$T = \frac{2 \cdot 3 \cdot 2 \cdot 10}{5} = 24 \text{ N}$$

Support: $2T = 48$ N

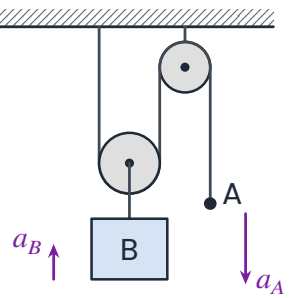
LAWS OF MOTION

Constraints • Springs • Lift

JEE & NEET PHYSICS

13 Constraint Relation

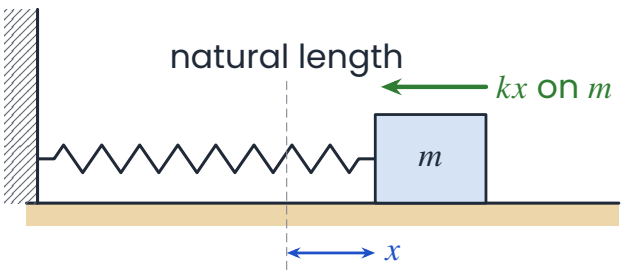
$$2y_B + y_A = \text{const} \Rightarrow a_B = \frac{a_A}{2}$$



Inextensible string, y from a fixed point. A goes down, B goes up.

14 Spring Force

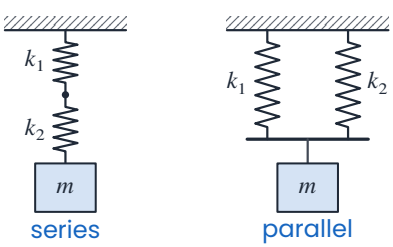
$$F = -kx$$



Ideal spring, x from natural length. Spring force cannot change suddenly; a string's tension can.

15 Springs in Series • Parallel

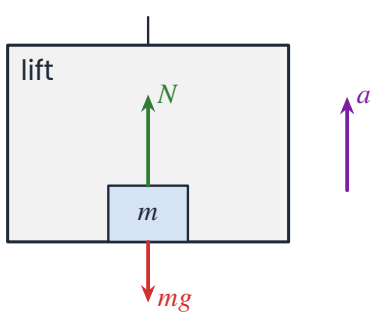
$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}, \qquad k_p = k_1 + k_2$$



Massless springs. Series: same force; parallel: same stretch. Cut in n equal parts: nk each.

16 Apparent Weight in a Lift

$$N = m(g \pm a)$$



$+$ for $\vec{a}$ up, $-$ for $\vec{a}$ down (a of the lift, not v). Free fall: $N = 0$, mass unchanged.

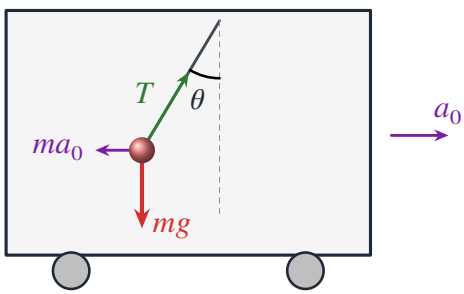
LAWS OF MOTION

Pseudo Force • Friction

JEE & NEET PHYSICS

17 Pseudo Force

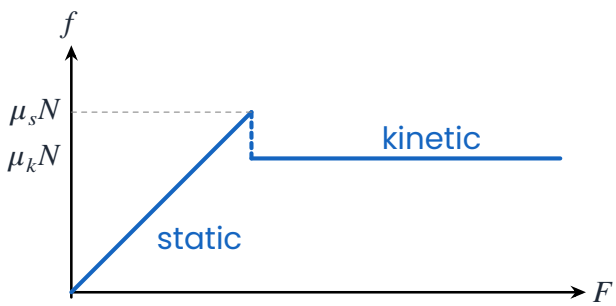
$$\vec{F}_{\text{pseudo}} = -m\vec{a}_0, \qquad \tan\theta = \frac{a_0}{g}$$



Used only in a frame accelerating at $\vec{a}_0$. θ: bob's tilt, at rest in the cart.

18 Static Friction

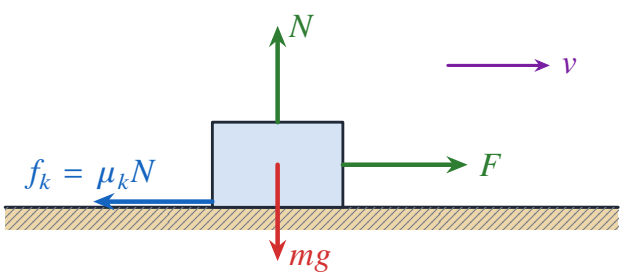
$$f_s \le \mu_s N, \qquad f_{s,\max} = \mu_s N$$



No sliding: f_s adjusts to the need (here $f = F$) up to $\mu_s N$; then $\mu_k N$ while sliding.

19 Kinetic Friction

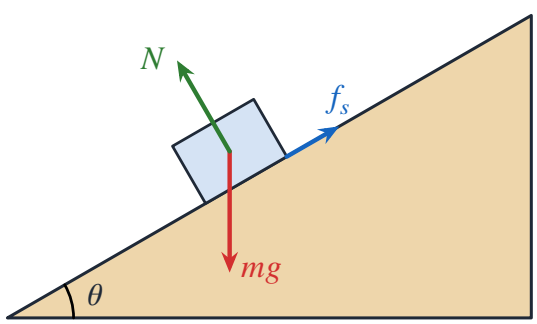
$$f_k = \mu_k N, \qquad \mu_k < \mu_s$$



While sliding: opposes the RELATIVE motion. N is not always mg.

20 Angle of Repose

$$\tan\theta_r = \mu_s$$



Block at rest, just about to slide on the incline: $f_s = mg\sin\theta$ is at its limit.

LAWS OF MOTION

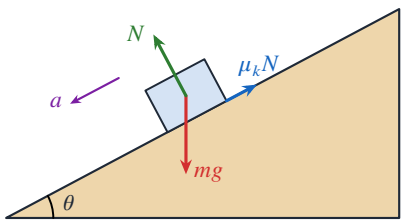
Friction • Inclines and Blocks

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21 Rough Incline

Down: $a = g(\sin\theta - \mu_k \cos\theta)$

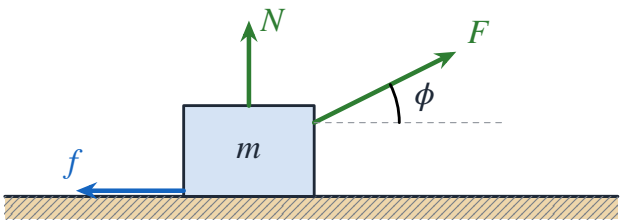
Up: $a = g(\sin\theta + \mu_k \cos\theta)$



While sliding down / up; both a point DOWN the slope. Slides down only if $\tan\theta > \mu_s$.

22 Least Force to Drag

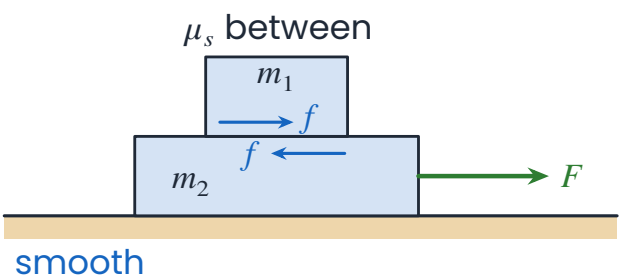
$$F_{\min} = \frac{\mu mg}{\sqrt{1+\mu^2}} \quad \text{at} \quad \tan\phi = \mu$$



Rough level floor, $\mu = \mu_s$ (just starts to move). Pulling up at ϕ reduces N.

23 Block on Block

$$F_{\max} = \mu_s(m_1 + m_2)g$$



Smooth floor, F on the lower block: both move together while $F \le F_{\max}$.

24 Quick Numerical: Incline

A block slides on a $37°$ incline, $\mu_k = 0.5$, $g = 10\ \text{m s}^{-2}$. Acceleration sliding down? Deceleration sliding up?

$$a_{\text{down}} = 10(0.6 - 0.4) = 2\ \text{m s}^{-2}$$

$$a_{\text{up}} = 10(0.6 + 0.4) = 10\ \text{m s}^{-2}$$

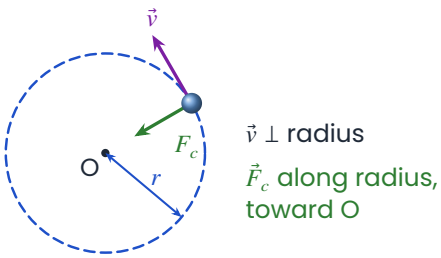
LAWS OF MOTION

Circular Motion • Roads

JEE & NEET PHYSICS

25 Centripetal Force

$$F_c = \frac{mv^2}{r} = m\omega^2 r \quad \text{(to centre)}$$



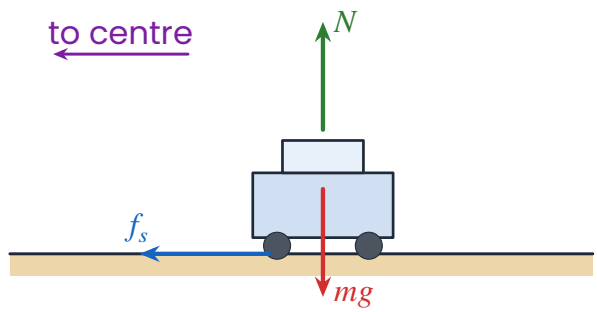
$\vec{v} \perp$ radius

$\vec{F}_c$ along radius, toward O

Uniform circular motion. It is the NET inward force (tension, friction, N...), not a new force.

26 Level Curved Road

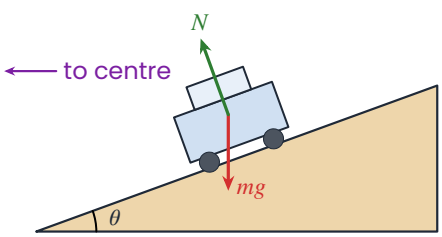
$$v_{max} = \sqrt{\mu_s r g}$$



Level road: static friction at the tyres supplies mv^2/r; it points to the centre.

27 Banked Road

$$v_{max} = \sqrt{rg\,\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$$



Friction acts down the slope at v_{max}. With $\mu_s = 0$: $\tan\theta = v^2/rg$ (no friction needed).

28 Quick Numerical: Roads

Curve of radius 40 m, $g = 10$ m s^{-2}. (a) Level, $\mu_s = 0.4$: v_{max}? (b) Banked, $\tan\theta = 0.25$: speed needing no friction?

$$v_{max} = \sqrt{0.4 \cdot 40 \cdot 10} \approx 12.6 \text{ m/s}$$

$$v = \sqrt{rg\tan\theta} = \sqrt{40 \cdot 10 \cdot 0.25} = 10 \text{ m/s}$$

LAWS OF MOTION

Quick Revision • Traps

JEE & NEET
PHYSICS

29 Every Result and When It Holds

Quantity	Result	Only if
Second law	$\vec{F} = d\vec{p}/dt = m\vec{a}$	inertial frame; $m\vec{a}$ for constant m
Momentum	$\vec{p}_{\text{total}}$ constant	$\sum \vec{F}_{\text{ext}} = 0$
Atwood	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$, $T = \frac{2m_1 m_2 g}{m_1 + m_2}$	light string, light smooth pulley
Lift	$N = m(g \pm a)$	+ for $\vec{a}$ up, − for $\vec{a}$ down
Friction	$f_s \le \mu_s N$, $f_k = \mu_k N$	static: no sliding; kinetic: sliding
Repose	$\tan\theta_r = \mu_s$	about to slide
Rough incline	$a = g(\sin\theta - \mu_k \cos\theta)$	sliding down
Level curve	$v_{\max} = \sqrt{\mu_s r g}$	level road, no skid
Banking	$\tan\theta = v^2/rg$	no friction needed

30 Common Traps

- Action–reaction act on different bodies: never cancel.
- $N = mg$ only on a level surface with no other vertical force and $a_y = 0$.
- Static friction is not always $\mu_s N$; that is only its limit.
- Friction opposes RELATIVE sliding; it can drive motion (top block).
- Centripetal force is not an extra force on the FBD.

31 Constants & Units

Quantity	Value
g near Earth's surface	$9.8\ \text{m s}^{-2}$ (≈ 10)
1 N	$1\ \text{kg m s}^{-2}$
Impulse, momentum	$\text{N s} = \text{kg m s}^{-1}$
$\sin 37^\circ,\ \cos 37^\circ$	$\approx 3/5,\ 4/5$
μ	no unit

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